

Lab task 6

1. Line Plot(Matplotlib)

Scenario:

You are tracking your water intake for a week to improve your health. You record how many glasses of water you drink each day and want to visualize your daily progress.

```
x = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun']  
y = [5, 6, 8, 7, 9, 10, 8]
```

2. Bar Chart

Scenario:

You run a small café and want to compare how many coffees were sold each day during the week to find your busiest day.

```
x = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun']  
y = [45, 60, 50, 70, 85, 95, 90]
```

3. Histogram

Scenario:

You conducted a small survey among your classmates to record how many hours they sleep at night. You want to see how the sleeping hours are distributed among the students.

```
sleep_hours = [5, 6, 7, 8, 5, 6, 7, 7, 8, 9, 5, 6, 8, 9, 7]
```

4. Scatter Plot

Scenario:

You are studying how much time students spend on social media versus their exam scores to see if more screen time affects their grades.

```
x = [1, 2, 3, 4, 5, 6, 7, 8] # Hours spent on social media  
y = [95, 90, 85, 80, 75, 70, 60, 55] # Exam scores
```

5. Pie Chart

Scenario:

You analyzed your monthly expenses and want to visualize how your money is divided among categories like Food, Rent, Transport, Shopping, and Savings.

```
categories = ['Food', 'Rent', 'Transport', 'Shopping', 'Savings']  
expenses = [25, 35, 15, 10, 15]
```

6. Area Plot

Scenario:

You want to visualize how your savings have grown month by month throughout the year to see your total progress visually.

'Month': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
'Savings': [2000, 2500, 2700, 3000, 3500, 4000]

7. 3D Plot

Scenario:

You are analyzing the relationship between students' **age**, **study hours**, and **exam scores** to understand how these three factors influence each other.

'Age': [18, 19, 20, 21, 22],
'Study_Hours': [2, 3, 4, 5, 6],
'Exam_Score': [60, 70, 75, 85, 90]

1. Bar Plot(seaborn)

Scenario:

You are a school teacher who wants to compare the **average marks** of students across different subjects to identify which subject needs more attention.

'Subject': ['Math', 'English', 'Science', 'History', 'Computer']
'Average_Score': [82, 75, 89, 70, 95]

2. Line Plot

Scenario:

You are managing a delivery company and want to track how your **monthly deliveries** increased or decreased over the first half of the year.

'Month': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
'Deliveries': [120, 140, 135, 160, 180, 200]

3. Histogram

Scenario:

You are a fitness trainer and want to see how your clients' **weights** are distributed to design personalized workout plans.

data = {'Weight': [55, 60, 62, 65, 70, 72, 75, 80, 85, 90, 95, 100]}